

Semi Conductor Memories



UNIT 3



- Semiconductor memory is used in any electronics assembly that uses computer processing technology.
- memory cards have become a common component for temporarily storing data - everything from the portable flash memory cards used for transferring files, to semiconductor memory cards used in cameras, mobile phones, etc

Semiconductor memory: main types



- To meet the growing needs for semiconductor memory, there are many types and technologies that are used.
- A variety of different memory technologies are available - each one suited to different applications.. Names such as ROM, RAM, EPROM, EEPROM, Flash memory, DRAM, SRAM, SDRAM, as well as F-RAM and MRAM are available, and new types are being developed to enable improved performance.
- Sram – static random access memory

FRAM vs MRAM



- *“FRAM or Ferroelectric Random Access Memory uses a ferroelectric capacitor architecture that employs ferroelectric materials as storage elements.*
- **The advantages of FeRam**
- FeRAM’s advantages over Flash include:
 - Lower power usage.
 - Faster write performance.
 - A much greater maximum read/write endurance
 - FeRAMs have data retention times of more than 10 years



- Why Replace a Ferroelectric RAM with an MRAM?

(magnetoeresistive random access memory)

- *“MRAM has unlimited performance and infinite Read/Write cycles; FRAM Reads are destructive and eventually lead to wear-out.*
- *MRAM Read/Write Cycle and Access times are faster, 35ns vs. 60ns access time,*
- *MRAM provides data retention for 20 years.*



- DDR3, DDR4, DDR5 and many more are seen and these refer to different types of SDRAM semiconductor memory
- Sdram – synchronous dynamic random access memory

ROM - Read Only Memory:



- A ROM is a form of semiconductor memory technology used where the data is written once and then not changed.
- Read-only memory, or ROM, is a type of computer storage containing non-volatile, permanent data.
- Almost every computer incorporates a small amount of ROM that contains the start-up firmware.



- This boot firmware is called the basic input/output system (BIOS). This software consists of code that instructs the boot-up processes for the computer -- such as loading the operating system (OS) into the random access memory (RAM)

Types of ROM



- **Programmable read-only memory-** PROM can be used to update firmware, such as BIOS, through the utilization of installation software.
- PROM is read-only memory that can be modified only once by a user. The user buys a blank PROM and enters the desired contents using a PROM program. Inside the PROM chip, there are small fuses which are burnt open during programming. It can be programmed only once and is not erasable.

Types of PROM include:



- **Ultraviolet-erasable ROM (UV-ROM)** - ROM whose contents can be erased using ultraviolet light, and then reprogrammed.
- **Erasable programmable ROM (EPROM)** - A type of ROM that is programmed using high voltages and exposure to ultraviolet light for about 20 minutes.



- **Electrically-erasable programmable ROM (EEPROM)** - Often used in older computer chips and to control BIOS, EEPROM can be erased and reprogrammed several times while enabling the erase and writing of only one location at a time.
- **Flash memory** is an updated version of EEPROM that allows numerous memory locations to be changed at the same time.

Advantages of ROM



- Non-volatile in nature
- Cannot be accidentally changed
- Cheaper than RAMs
- More reliable than RAMs
- Static and do not require refreshing
- Contents are always known and can be verified

RAM - Random Access Memory



- As the names suggest, the RAM or random access memory is a form of semiconductor memory technology that is used for reading and writing data in any order - in other words as it is required by the processor.



- Your computer RAM is essentially short term memory where data is stored as the processor needs it.
- This isn't to be confused with long-term data that's stored on your hard drive, which stays there even when your computer is turned off.
- Anytime you play a game from your computer's hard drive or stream a movie from the Internet, all of the data that your computer's processor needs to play the game or watch the movie is stored in RAM. This is so that your processor can get to it quickly.



- Once you're done playing your game or watching your movie and you close it down, the processor doesn't need that data anymore, so it replaces it with the next task you give it. RAM can slow down your computer if there isn't enough of it for the processor to perform the tasks you ask it to



- Many types of RAM including SDRAM with its DDR3, DDR4, and soon DDR5 variants are used in huge quantities.
- Ddr – double data rate synchronous dynamic random access memory



SDRAM

It refers as synchronous dynamic random access memory

SDRAM was released in 1997

SDRAM has less speed in comparison DDR

SDRAM speed is considered as 66 MHz, 100 MHz, 133 MHz

DDR

it refers as Double data rate SDRAM

DDR RAM was released in 2000

DDR can transfer data at roughly twice the speed of SDRAM.

DDR working on speed as 200 MHz, 266 MHz, 333 MHz, 400 MHz



END